Contal and O’Quigley Method

By Jeff Fine

Determining Cut-Points

- Often, we get CTSC projects where we get an exploratory continuous variable and are asked to find the best cut point to make it categorical.
  - Very popular is Youdin's Index (optimizing sensitivity and specificity)

- However, for survival analyses, Youdin's Index does not consider censoring. Introducing the Contal and O'Quigley Method (1999)

Math

- The Contal and O'Quigley method looks to find the optimal cutoff point using the log rank test statistic

$$\text{Log Rank Statistic} = L_k(t) = \sum_{i=1}^{k} \left(d_i^+ - d_i \frac{r_i^+}{r_i} \right)$$

- If we define R to be our risk factor of interest (continuous) and T be the outcome variable (i.e. death) then we can let $t_{(1)} < t_{(2)} < \ldots < t_{(k)}$ be the K distinct values of ordered observed event times of the outcome variable T. We will also define C be the set of K distinct values of the continuous risk factor R.
- Therefore, based on a hypothetical cut-point for C, d_i would be the number of events at time $t_{(i)}$, r_i be the number of subjects at risk prior to time $t_{(i)}$ and d_i^+ and r_i^+ be the number of events at time $t_{(i)}$ in group R > C and number of subjects at risk just prior to $t_{(i)}$ in the group R > C.
- Optimal cut-point is that the value of C, C_k, that maximizes the absolute value of $L_k(t)$

Math continued

- Therefore, C_k gives the value of the continuous risk factor that gives the maximum difference between the subjects in the two groups defined by the cut-point.
- Contal and O'Quigley also recommend to test the significance of the cut-point by using this test statistic (Q):

$$\text{Test Statistic} = q = \frac{1}{s\sqrt{k-1}} \max \left| L_k(t) \right| \qquad \text{where, } s^2 = \frac{1}{(k-1)} \sum_{j=1}^{k} a_j^2$$

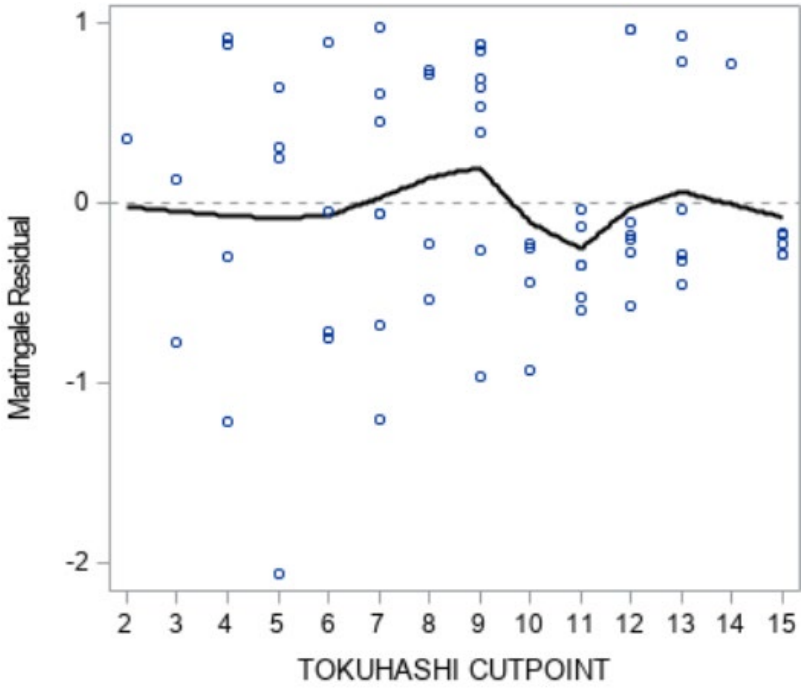
a_j's are the scores associated with the jth death, given by $1 - \sum_{i=1}^{j} \frac{1}{k - i + 1}$

Example from CTSC

- Retrospective chart review of patients undergoing surgery for metastatic spine lesions. Data was extracted from the electronic medical records of 65 patients who underwent surgery at UC Davis Health in the past 7 years.
- Primary Hypothesis: understanding if there is a difference in common risk factors (i.e. race, insurance status, sex, etc.) and time to event (death).
- Secondary Hypothesis: is there an optimal cut-off for the Tokuhashi scoring system (scoring system for the preoperative evaluation of a patient's prognosis with a metastatic spinal tumor) that could be used to best predict time to event. The Tokuhashi scoring system goes from 0-15 where the higher the score, the better the prognosis.

Is a cutoff needed?

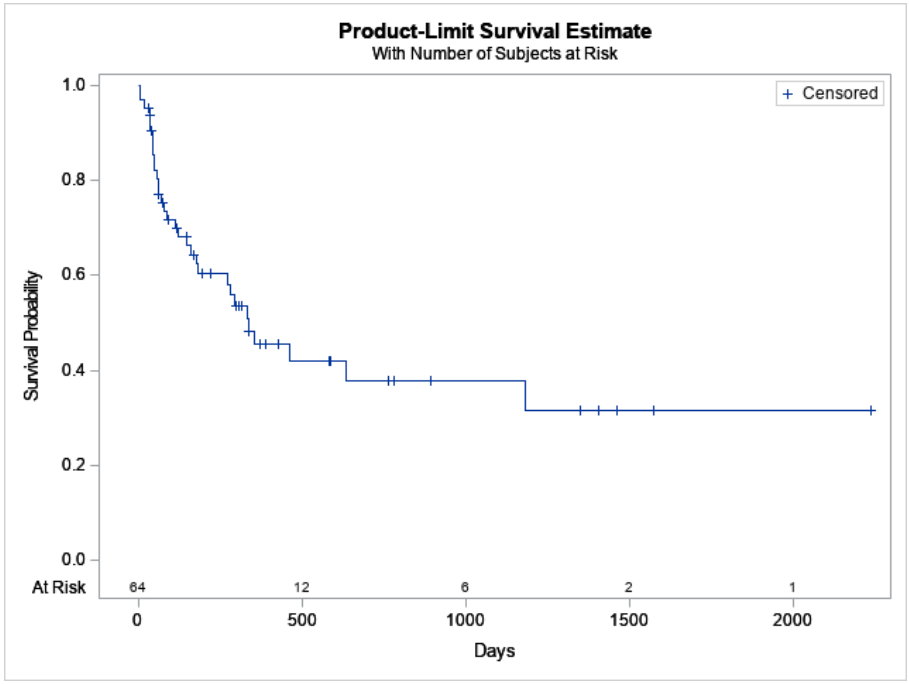
- We can use a plot of the cutoff points and martingale residuals to see if a cutoff is needed and where the best place might be
- Based on the figure, it looks like treating Tokuhashi score as linear would be inappropriate.
- Furthermore, it looks like a Tokuhashi cutoff point of 8/9 or 10/11 would work best.



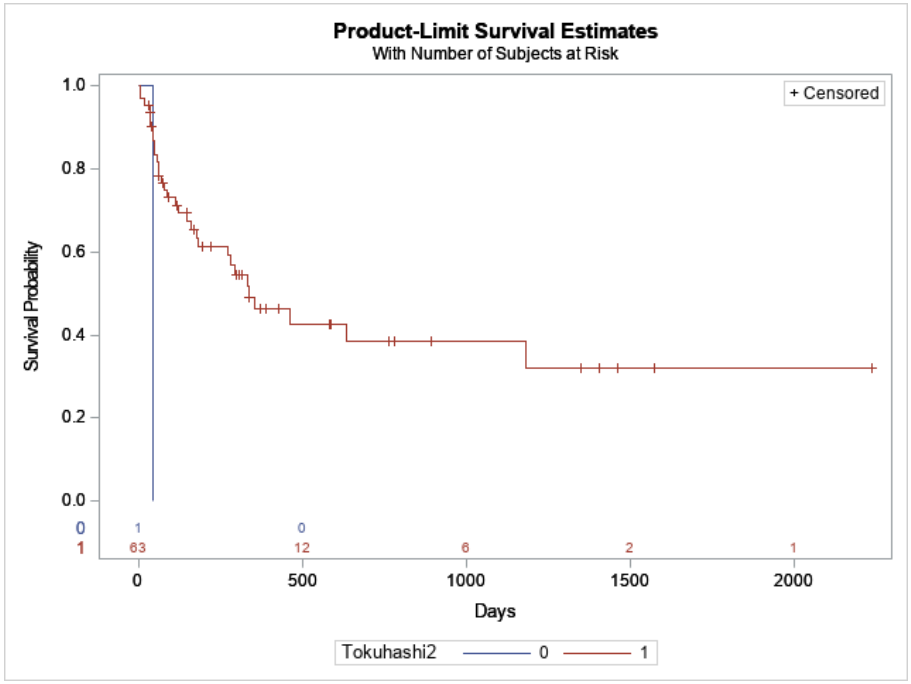
Let's Calculate the Log Rank Statistic by Hand

- Let's do a quick example with this dataset where we use a cutoff of 3 for the Tokuhashi score:

Full dataset:



Cutoff of 3:



Calculation continued

Everyone								Tokuhashi Score > 3								
Days		Survival	Failure	Survival Standard Error	Number Failed	Number Left		Days		Survival	Failure	Survival Standard Error	Number Failed	Number Left		Log rank for each time point
0		1	0	0	0	64		0		1	0	0	0	63		0
3		0.9844	0.0156	0.0155	1	63		3		0.9841	0.0159	0.0157	1	62		0.015625
5		0.9688	0.0313	0.0217	2	62		5		0.9683	0.0317	0.0221	2	61		0.015873
18		0.9531	0.0469	0.0264	3	61		18		0.9524	0.0476	0.0268	3	60		0.016129
31	*	.	.	.	3	60		31	*	.	.	.	3	59		0
31	*	.	.	.	3	59		31	*	.	.	.	3	58		0
33		0.937	0.063	0.0305	4	58		33		0.936	0.064	0.031	4	57		0.016949
35	*	.	.	.	4	57		35	*	.	.	.	4	56		0
36		0.9205	0.0795	0.0341	5	56		36		0.9192	0.0808	0.0346	5	55		0.017857
37		0.9041	0.0959	0.0373	6	55		37		0.9025	0.0975	0.0378	6	54		0.017857
39	*	.	.	.	6	54		39	*	.	.	.	6	53		0
44		0.8539	0.1461	0.0451	9	51								51		-0.94444
46		0.8371	0.1629	0.0472	10	50		46		0.8514	0.1486	0.0458	9	50		0
50		0.8204	0.1796	0.0491	11	49		50		0.8344	0.1656	0.0479	10	49		0
55		0.8036	0.1964	0.0509	12	48		55		0.8174	0.1826	0.0499	11	48		0
59		.	.	.	13	47		59		.	.	.	12	47		0

|Sum of Log Rank Statistic| = 0.844

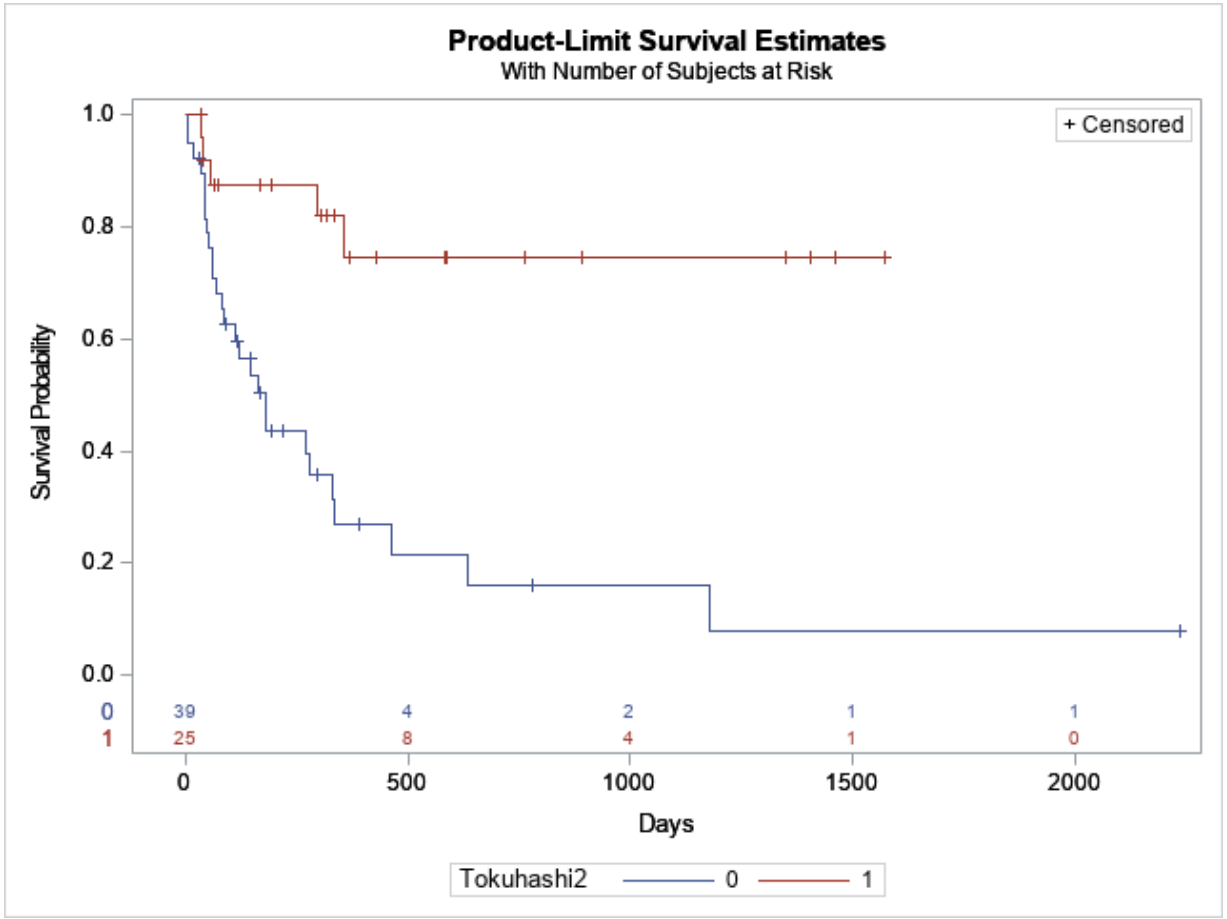
Output in SAS:

Cut-Points		Contal and O'Quigley Method					Cox Model Hazard Ratio		Cox Model Wald P-value		False Discovery Rate	
Cut Level	TOKUHASHI	SK	Absolute SK	Q Statistic	P-value	Selected Cut-Point	Hazard Ratio	Selected Cut-Point	P-value	Selected Cut-Point	P-value	Selected Cut-Point
1	1	0	0	0	0.3000		.		.		.	
2	2	0	0	0	0.3000		0.141		0.0644		0.0698141	
3	3	-0.844467	0.8444672	0.1687641	0.3000		0.272		0.0367		0.0477062	
4	4	-2.089576	2.0895758	0.417595	0.3000		0.249		0.0016		0.002896	
5	5	-4.829226	4.8292265	0.9651054	0.3000		0.278		0.0007		0.001534	
6	6	-6.85786	6.8578604	1.3705214	0.0467		0.296	<====	0.0007		0.001534	
7	7	-7.934389	7.9343888	1.5856621	0.0131		0.293		0.0006		0.001534	
8	8	-8.842914	8.8429139	1.767228	0.0039		0.270		0.0004		0.001534	
9	9	-9.857032	9.8570319	1.9698962	0.0009		0.131		<.0001	<====	0.0005374	<====
10	10	-13.0629	13.062896	2.6105778	<.0001	<====	0.185		0.0006		0.001534	
11	11	-10.60218	10.60218	2.1188116	0.0003		0.287		0.0114		0.0185293	
12	12	-7.250933	7.2509331	1.4490757	0.0300		0.247		0.0227		0.0327489	
13	13	-6.12616	6.1261601	1.2242934	0.0998		0.151		0.0640		0.0698141	
14	14	-4.453057	4.4530575	0.8899292	0.3000		0.000		0.9911		0.9910907	
15	15	-4.678428	4.6784277	0.9349688	0.3000		.		.		.	

Using the SAS MACRO (FindCut_Macro.SAS) we get the same result for cut level 3 for Tokuhashi. The SAS output says that the largest Absolute SK value is 13.062896 and so a cut-off of 10 should be used. This is also statistically significant (p-value < 0.0001). This suggests that the cut-point obtained is significant and that Tokuhashi is related to time to event (death).

Kaplan Meier Curve with Tokuhashi (binary)

Test of Equality over Strata			
Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	14.6466	1	0.0001
Wilcoxon	9.6741	1	0.0019
-2Log(LR)	21.6407	1	<.0001



Limitations

- Only good for dichotomization
- Loss of information when categorizing a continuous variable (i.e. Possible loss of power)
- Potential confounding that might arise from categorization and using open-ended categories

Package for R

- https://search.r-project.org/CRAN/refmans/survMisc/html/cutp.html

References:

- https://support.sas.com/resources/papers/proceedings/proceedings/sugi28/261-28.pdf
- https://www.sciencedirect.com/science/article/pii/S0167947398000966
- https://support.sas.com/resources/papers/proceedings15/3249-2015.pdf
- SAS macro download: http://bioinformaticstools.mayo.edu/research/findcut/

Discussion (problems I ran into):

- Calculating the Log Rank Statistic when using a cutoff point of 3 worked okay but when trying to do it by hand for a cutoff point of 10 resulted in a different answer than what SAS Macro gives.
- I believe the equation given is when there is no censoring. Taken from Contal and O'Quigley paper (1999):

where d_i is the number of deaths at time $x_{(i)}$, r_i is the number of patients at risk just before $x_{(i)}$, and the indices $^+$ and $^-$ denote the number in each group defined by the hypothetical cutpoint μ. The log-rank statistic, for some fixed μ is

$$U=\sum_{i=1}^{k} d_i^+ - d_i r_i^+ r_i .$$

In the case of no censoring, $k=n$, the log-rank statistic is exactly the Savage test statistic, i.e. the linear rank statistic with exponential scores. The associated scores are defined by:

$$a_i = 1 - \sum_{j=1}^{i} \frac{1}{k-j+1}$$

for the ith death. The expectation of the scores $\bar{a}$ is 0, and its variance σ^2 is $[1/(k-1)]\sum_{j=1}^{k} a_j^2$. The process $S_n(t)$ is identical to

$$\frac{1}{\sigma k-1}\sum_{i=1}^{[kt]} a_i = \frac{1}{\sigma k-1}\sum_{i=1}^{k} d_i^+ - d_i r_i^+ r_i .$$

In the case of censoring, $k \neq n$, the log-rank statistic is used, and normalised by the variance of the Savage test scores on the deceased patients.

$$S_n(t) = \frac{1}{\sigma k-1}\sum_{i=1}^{k} d_i^+ - d_i r_i^+ r_i ,$$

where $\sigma^2=[1/(k-1)]\sum_{j=1}^{k} a_j^2$. Under the assumption of a random censorship model, the process has the same asymptotic distribution as the Brownian bridge.

- findcut(data=survival2, time=Days, event=Death2, cutvar=Tokuhashi, plotop=1,residop=1, cutpoints=1 to 15 by 1);